Financial Time Series Analysis

Lecture 3

- TA Session Friday April 2, 2021, 11:00am - 12:00pm (EDT)

Outline

- Autocovariance/autocorrelation functions
  - Bartlett's formula
  - Ljung-Box test
- Basic time series models
  - Autoregressive Processes (AR)
  - Partial autocorrelation function (PACF)

Autocorrelation Function: 1

- The autocovariance and autocorrelation functions are central to time series analysis. They form the basis of time domain analysis.
- The autocovariance and autocorrelations are Fourier transforms of spectral density functions (over frequencies). These form the basis of frequency domain analysis.
- While frequency domain analysis is widely used, especially in engineering applications, it is very infrequently used in economics and finance. We pursue time domain analysis in this course, but the Course Notes will have a section on this approach.

Autocorrelation Function: 2

- We will want to estimate this function from data.
- The sample autocovariance function is computed in exactly the same way as a sample covariance function:

$$\hat{\gamma}(h) = \frac{1}{n} \sum_{t=1}^{n-h} (X_{t+h} - \bar{X})(X_t - \bar{X}).$$

- Some textbooks (and software packages) use $n - h$ as the divisor rather than n. $n - h$ gives an unbiased estimator.
- The denominator (n or $n - h$) is a parameter in statsmodels.tsa.stattools.acovf (unbiased = True) gives $n - k$, (False gives n), the default is False.
- Same for statsmodels.tsa.stattools.acf
-
$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}.$$

Autocorrelation Function: 3

- We are concerned with the structure/shape of the autocorrelation/autocorrelation function. The shape can help to identify the underlying process.
- First,we ask if the autocorrelation is consistent with an independent process with mean 0 and variance σ^2. That is, at any lag h, we want to test whether $\rho(h) = 0, h \neq 0$ using $\hat{\rho}(h)$.
- The problem of testing for the presence of autocorrelation at a particular lag, h, was first addressed by M. Bartlett in 1946, and he derived "Bartlett's formula." This formula gives the asymptotic behavior of $\hat{\rho}(h)$ for large sample sizes, and it is the source of the parallel lines denoting the 95% confidence intervals in ACF plots.
- There is a "rule of thumb" concerning the amount of data needed in order to estimate $\rho(h)$. Specifically, $n \geq \max(4h, 50)$.

Bartlett's Formula: 1

Following Brockwell and Davis:

- Bartlett assumed $X_t - \mu = \sum_{j=-\infty}^{\infty} \psi_j Z_{t-j}$ where $\sum_{j=-\infty}^{\infty} |\psi_j| < \infty$, $\{Z_t\}$ iid, and $E(Z_t^4) < \infty$, then

$$\sqrt{n}(\hat{\boldsymbol{\rho}}(h) - \boldsymbol{\rho}(h)) \Rightarrow N_h(\mathbf{0}, \mathbf{W})),$$

where

$$\hat{\boldsymbol{\rho}} = (\hat{\rho}(1), \cdots, \hat{\rho}(h)),$$

$$\boldsymbol{\rho} = (\rho(1), \cdots, \rho(h))$$

and **W** is a covariance matrix whose elements are given by

Bartlett's Formula: 2

Bartlett's formula:

$$
\begin{aligned}
w_{ij} &= \sum_{k=-\infty}^{\infty} (\rho(k+i)\rho(k+j) + \rho(k-i)\rho(k+j) \\
&+ 2\rho(i)\rho(j)\rho^2(k) - 2\rho(i)\rho(k)\rho(k+j) \\
&- 2\rho(j)\rho(k)\rho(k+i))
\end{aligned}
$$

- For a given model, e.g. AR(1), $\gamma(h)$ can be determined, hence the correlation structure of the empirical correlation coefficients can be determined.
- For example, suppose, we want to test whether or not $\rho(h) = 0$ if $h \neq 0$, i.e. the process is white noise.

Bartlett's Formula: 3

- Consider the asymptotic distribution of $\sqrt{n}(\hat{\rho}(h) - 0)$. From Bartlett's formula, $w_{hh} = 1$ $(i = j = 1)$. The first term is 1, while all the others are 0. Hence, if the process is white noise,
$$\sqrt{n}(\hat{\rho}(h) - 0) \Rightarrow N(0, 1).$$
- For a 95% confidence interval, the confidence limits for $\rho(h)$ are $\pm 1.96\sqrt{n}$.
- The w_{ij} can be calculated for some standard time series models, for example for an AR(1), MA(q) etc. We will display those formulas later.

Confidence Intervals for $\hat{\rho}_k$

- Suppose we have a AR(1) process with $\rho(h) = \phi_1^{|h|}$.
- One can apply Bartlett's formula to find the asymptotic distribution for $\hat{\rho}(h)$,

$$\sqrt{n}(\hat{\rho}(h) - \rho(h)) \Rightarrow N(0, w_{hh}),$$

where

$$w_{hh} = 1 + 2\sum_{i=1}^{h-1} \rho(i)^2.$$

Ljung-Box Test: 1

- A shortcoming of the Bartlett approach is that we are testing many separate hypotheses simultaneous, checking to see whether the sample autocorrelation plot lies within the upper and lower limit. If so, we conclude that the process (residuals) are white noise.
- The difficulty with this is that we are conducting simultaneous statistical tests, each at a particular α level (usually .05) and making a simultaneous judment about all the tests. Instead, we need a test that will keep the experimentwise error rate to the desired level (usually .95.

Ljung-Box Test: 2

- A simultaneous test of whether a sample autocorrelation function comes from an iid white noise process was first developed by Box and Pierce in 1970. They developed a test for the composite null hypothesis;
  $H_0 : \rho(1) = \rho(2) = \cdots = \rho(m) = 0$
  $H_1 :$ at least one of $\rho(i),\ 1 \leq i \leq m$ is not 0.
- For a data set of size n, the test statistics was

$$Q^*(m) = n \sum_{i=1}^{m} \hat{\rho}^2(i).$$

- Under H_0 the test statistics had an asymptotic χ^2_m distribution.

Ljung-Box Test: 3

- In 1978, Ljung and Box published an improved test that had greater power in finite samples. The test statistic was

$$Q(m) = n(n+2)\sum_{i=1}^{m} \frac{\hat{\rho}_i^2}{n-i},$$

  and was tested using the same χ_m^2 distribution.
- This can be called using Python statsmodels

  statsmodels.stats.diagnostic.acorr_ljungbox(x,lags)

  with other parameters available.

Ljung-Box Test: 4

IID Normal N(0,1) errors, n = 1,000

import statsmodels as smm

smm.stats.diagnostic.acorr_ljungbox(sim, lags = 10)

- (array([0.18728497, 3.07830832, 3.12318292, 4.65185238, 8.69101277, 10.48872061, 11.11405683, 12.85301371, 12.89994, 12.9098376]),
- array([0.66518598, 0.21456251, 0.37301997, 0.32491944, 0.12204167, 0.10552291, 0.13372072, 0.11700801, 0.16718639, 0.22875669]))

Ljung-Box Test: 5

AR(1), $\phi_1 = .2$, IID N(0,1) errors, n = 200

import statsmodels as smm

smm.stats.diagnostic.acorr_ljungbox(x, lags = 10)

- (array([6.64749532, 6.94627588, 7.08514479, 8.27915957, 11.64544475, 11.66216157, 11.66359782, 15.4305607 , 19.57099227, 20.5614858]),
- (array([0.00992953, 0.03101954, 0.06923287, 0.08187116, 0.03998259, 0.06994412, 0.1121786 , 0.05129471, 0.02075263, 0.02436759]))

Ljung-Box Test: 6

AR(1), $\phi_1 = .2$, IID N(0,1) errors, n = 100

import statsmodels as smm

smm.stats.diagnostic.acorr_ljungbox(x, lags = 10)

- array([2.52656454, 4.05288395, 4.52154401, 5.31449319, 7.28748766, 7.51645729, 7.81052663, 7.90728538, 8.36476559, 8.36484189]),
- array([0.1119437 , 0.13180365, 0.21037664, 0.25652324, 0.20012234, 0.27571037, 0.34959777, 0.44257887, 0.4978452 , 0.5932462])

Autoregressive Processes

Autoregressive Processes: 1

- An autoregressive process, $\{X_t,\ t \geq 0\}$, is a stochastic model in which the process is regressed on the lagged values of itself plus an additive noise term, so AR(p) model is given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t,$$

  where $\{\epsilon_t,\ t \geq 0\}$ is a $\text{WN}(0, \sigma_\epsilon^2)$.
- There are a number of issues to address including
  - Whether the X_t process is weakly stationary,
  - Determining the moments of the X_t process, especially $\gamma(h)$
  - Identifying the order of an AR process from data
  - Parameter estimation
  - Predicting future values of an AR process.
- We start with AR(1)

Autoregressive Processes: 2

We begin with the AR(1) process:

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t,$$

where the ϵ_t process is $\mathrm{WN}(0, \sigma^2)$.

- We iterate the above expression backward to time 0 (or $-\infty$) depending upon the index set)

$$\begin{aligned}
X_t &= \phi_0 + \phi_1 X_{t-1} + \epsilon_t \\
&= \phi_0 + \phi_1(\phi_0 + \phi_1 X_{t-2} + \epsilon_{t-1}) + \epsilon_t \\
&= \phi_0(1 + \phi_1) + \phi_1^2 X_{t-2} + \sum_{i=0}^{1} \phi_1^i \epsilon_{t-i} \\
&= \phi_0(1 + \phi_1 + \phi_1^2) + \phi_1^3 X_{t-3} + \sum_{i=0}^{2} \phi_1^i \epsilon_{t-i} \\
&= \phi_0 \sum_{i=0}^{t-1} \phi_1^i + \phi_1^t X_0 + \sum_{i=0}^{t-1} \phi_1^i \epsilon_{t-i}
\end{aligned}$$

Autoregressive Processes: 3

Assuming the index set is $\{t \geq 0\}$:

$$X_t = \begin{cases} \phi_0 \frac{1-\phi_1^t}{1-\phi_1} + \phi_1^t X_0 + \sum_{i=0}^{t-1} \phi_1^{i-1} \epsilon_{t-i} & \text{if } |\phi_1| < 1, \\ \phi_0 t + X_0 + \sum_{i=0}^{t} \epsilon_{t-i} & \text{if } |\phi_1| = 1 \ , \\ \phi_0 \frac{\phi_1^t - 1}{\phi_1 - 1} + \phi_1^t X_0 + \sum_{i=0}^{t-1} \phi_1^{i} \epsilon_{t-i} & \text{if } |\phi_1| > 1 \ , \end{cases}$$

- It is clear with the above expressions, if $|\phi_1| > 1$, X_t will explode.
- If $\phi_1 = 1$ then the process is a random walk and the variance will not be constant (see later), then the process will not be stationary.

Autoregressive Processes: 4

- It is clear that $|\phi_1| < 1$ is a necessary condition for the X_t to be weakly stationary, whether the index set is $\{t \geq 0\}$ or $\{-\infty < t < \infty\}$.
- For weak stationarity we need the mean and variance to be constant as well as an appropriate covariance function. Defining $E(X_t) = m_t$ and recalling $E(\epsilon_k) = 0$.

$$
\begin{aligned}
m_t &= \phi_0 + \phi_1 m_{t-1} + 0 \\
&= \phi_0(1 + \phi_1) + \phi_1^2 m_{t-2} \\
&\cdots \\
&= \phi_0 \frac{1 - \phi_1^t}{1 - \phi_1} + \phi_1^t m_0
\end{aligned}
$$

Autoregressive Processes: 5

- Interestingly, if we take $m_0 = \phi_0/(1-\phi_1)$, then $m_t = m_0 = \phi_0/(1-\phi_1), \ \forall t$.
- Thus the constant mean property is fulfilled if we initialize $E(X_0) = \phi_0/(1-\phi_1)$. If we start with a different initial value and $|\phi_1| < 1$, then the process is asymptotically stationary, provided the variance and covariance are appropriate.
- We can use the same iteration procedure for the variance, Letting $Var(X_t) = \sigma_t^2$,

Autoregressive Processes: 6

$$
\begin{aligned}
\sigma_t^2 &= Var(\phi_0 + \phi_1 X_{t-1} + \epsilon_t) \\
&= \phi_1^2 \sigma_{t-1}^2 + \sigma_\epsilon^2 \\
&= \phi_1^4 \sigma_{t-2}^2 + \sigma_\epsilon^2 (1 + \phi_1^2) \\
&\ldots \\
&= \phi_1^{2t} \sigma_0^2 + \sigma_\epsilon^2 \sum_{i=0}^{t-1} \phi_1^{2i}
\end{aligned}
$$

Thus

$$
\sigma_t^2 = \begin{cases} \sigma_\epsilon^2 \frac{1-\phi_1^{2t}}{1-\phi_1^2} + \phi_1^{2t} \sigma_0^2 & \text{if } |\phi_1| < 1, \\ t\sigma_\epsilon^2 + \sigma_0^2 & \text{if } |\phi_1| = 1\ , \\ \sigma_\epsilon^2 \frac{\phi_1^{2t}-1}{\phi_1^2-1} + \phi_1^{2t} \sigma_0^2 & \text{if } |\phi_1| > 1\ , \end{cases}
$$

Autoregressive Processes: 7

We now assume that $|\phi_1| < 1$, a necessary condition for weak stationarity of an AR(1) process.

- Interestingly, if we let $\sigma_0^2 = \sigma_\epsilon^2/(1 - \phi_1^2)$, then

$$Var(X_t) = \sigma_t^2 = \sigma_\epsilon^2/(1 - \phi_1^2),\ \forall t \geq 0.$$

- We next calculate the autocovariance function for an AR(1) process, assuming $|\phi_1| < 1$, $E(X_0) = \phi_0/(1 - \phi_1)$ and $Var(X_0) = \sigma^2/(1 - \phi_1^2)$

Autoregressive Processes: 8

$$
\begin{aligned}
Cov(X_t, X_{t+h}) &= Cov(\sum_{i=0}^{t} \phi_1^i \epsilon_{t-i}, \sum_{i=0}^{t+h} \phi_1^i \epsilon_{t+h-i}) \\
&= Cov(\sum_{i=0}^{t} \phi_1^i \epsilon_{t-i}, \sum_{i=0}^{h-1} \phi_1^i \epsilon_{t+h-i} + \sum_{i=h}^{t+h} \phi_1^i \epsilon_{t+h-i}) \\
&= Cov(\sum_{i=0}^{t} \phi_1^i \epsilon_{t-i}, \sum_{i=0}^{h-1} \phi_1^i \epsilon_{t+h-i} + \phi_1^h \sum_{i=0}^{t} \phi_1^i \epsilon_{t-i}) \\
&= 0 + \phi_1^h Var(X_t) \\
&= \phi_1^h \frac{\sigma_\epsilon^2}{1 - \phi_1^2}
\end{aligned}
$$

Autoregressive Processes: 9

We summarize properties of an AR(1) process $\{X_t,\ t \geq 0\}$

$$X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t,$$

where $\{\epsilon_t,\ t \geq 0\}$ is WN$(0, \sigma_\epsilon^2)$, $E(X_0) = \phi_0/(1-\phi_1)$, and $Var(X_0) = \sigma_\epsilon^2/(1-\phi_1^2)$, $|\phi_1| < 1$

- The X_t process is weakly stationary if and only if $|\phi_1| < 1$.
- $E(X_t) = \phi_0/(1-\phi_1)$,
- $Var(X_t) = \sigma_\epsilon^2/(1-\phi_1^2)$, $|\phi_1| < 1$
- $\gamma(h) = \phi_1^{|h|} \frac{\sigma_\epsilon^2}{1-\phi_1^2}$, $|\phi_1| < 1$
- $\rho(h) = \phi_1^{|h|}$, $|\phi_1| < 1$

Autoregressive Processes: 10

- Note, if either $E(X_0)$ or $Var(X_0)$ differ from the stationary values, but $|\phi_1| < 1$, then the process is not weakly stationary (because the mean and variance are not constant); however, it is asymptotically stationary.
- Sometimes, it is convenient to remove the mean. We can do that by defining $\mu = \phi_0/(1 - \phi_1)$ and then defining $Y_t = X_t - \mu$. The resulting sequence $\{Y_t,\ t \geq 0\}$ now has mean 0.

Causal Processes

- As an aside, we have a representation of a weakly stationary process X_t as

$$X_t = \sum_{i=0}^{t} \phi_1^i \epsilon_{t-i}, \; t \geq 0, \text{ or}$$

$$X_t = \sum_{i=0}^{\infty} \phi_1^i \epsilon_{t-i}, \; -\infty < t < \infty$$

- More generally, if

$$X_t = \sum_{i=0}^{\infty} \psi_i \epsilon_{t-i},$$

  satisfying $\sum_{i=0}^{\infty} |\psi_i| < \infty$, then the process is called causal.

- Causality is the same concept as non-anticipate that arises in stochastic calculus, "not looking into the future."

AR(2) Processes: 1

- We now consider AR(2) processes:

$$X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t,$$

  for a linear regression model whose explanatory (independent) variables are the first two lagged values of the time series.
- Rather than repeating the iterations as we did before and then determining the mean, variance and autocovariance, we assume that the moment structure is constant and work with the definition of X_t.
- Taking expectations, and assuming that $E(X_t) = \mu, \ \forall t$, we find $\mu = \phi_0 + \mu(\phi_1 + \phi_2) + 0$, or $\mu = \phi_0/(1 - \phi_1 - \phi_2)$, assuming $\phi_1 + \phi_2 \neq 1$.

AR(2) Processes: 2

- By substituting out ϕ_0 and defining $Y_t = X_t - \mu$, we find

$$Y_{t+h} = \phi_1 Y_{t+h-1} + \phi_2 Y_{Y+h-2} + \epsilon_{t+h}$$

- Multiplying each term by (Y_t) and taking expectations, we find

$$\gamma(h) = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}, h > 0$$

  because the covariance of ϵ_{t+h} and Y_t is 0 and the means are 0.

- Setting h = 1, recognizing that $\gamma(h) = \gamma(-h)$, and dividing by $\gamma(0)$ to obtain the autocorrelation,

$$\rho(h) = \phi_1 \rho(h-1) + \phi_2 \rho(h-2),$$

  and for $h = 1$, $\rho(1) = \phi_1 \rho(0) + \phi_2 \rho(1)$

AR(2) Processes: 3

- Since $\rho(0) = 1$, we can find the AR(2) autocorrelation function

$$\rho(h) = \begin{cases} 1 & \text{if } h = 0, \\ \phi_1/(1-\phi_2) & \text{if } h = 1, \\ \phi_1\rho(h-1) + \phi_2\rho(h-2) & \text{if } h \geq 2. \end{cases}$$

- If we define the backshift operator B, $BX_t = X_{t-1}, B^2X_{t-2}, \cdots$. Using this, the autocorrelation function satisfies the a second order difference equation:

$$(1 - \phi_1 B - \phi_2 B^2)\rho(h) = 0.$$

- We will see that the roots of the associated quadratic equation

$$1 - \phi_1 x - \phi_2 x^2 = 0,$$

as being central to conditions on ϕ_1, ϕ_2 to have weak stationarity.

AR(2) Processes: 4

- Consider an AR(2) process, $\{X_t, \ -\infty < t < \infty\}$:

$$X_t = \phi_0 + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \epsilon_t,$$

  where $\{\epsilon_t\}$ is $WN(0, \sigma^2)$.
- We want to determine conditions under which the X_t process is weakly stationary.
- We convert this to a vector process for $(X_t, X_{t-1})^T$ using the defining equation give above for the X_t component and $X_{t-1} = X_{t-1}$ for the X_{t-1} component.
- This gives the following representation:

$$\mathbf{X}_t = \boldsymbol{\mu} + \mathbf{M}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t,$$

  where $\boldsymbol{\mu} = (\phi_0, 0)^T$, $\boldsymbol{\epsilon}_t = (\epsilon_t, 0)^T$.

AR(2) Processes: 5

Finally,

$$\mathbf{M} = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix}.$$

One can iterate backward as before:

$$\begin{aligned} \mathbf{X}_t &= \boldsymbol{\mu} + \mathbf{M}\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t \\ &= \boldsymbol{\mu} + \mathbf{M}\left(\boldsymbol{\mu} + \mathbf{M}\mathbf{X}_{t-2} + \boldsymbol{\epsilon}_{t-1}\right) + \boldsymbol{\epsilon}_t \\ &= \cdots \\ &= \sum_{i=0}^{t-1} \mathbf{M}^i \boldsymbol{\mu} + \mathbf{M}^t \mathbf{X}_0 + \sum_{i=0}^{t-1} \mathbf{M}^i \boldsymbol{\epsilon}_{t-i} \end{aligned}$$

AR(2) Processes: 6

- To ensure stationarity, we need the matrix $\mathbf{M}^t \to \mathbf{0}$ as $t \to \infty$.
- Consequently, we need the two eigenvalues of this matrix to converge to 0.
- The eigenvalues are the roots of the determinantal equation:

$$(\phi_1 - \lambda)(0 - \lambda) - (\phi_2)(-1) = 0.$$

- Writing this as a quadratic equation in λ

$$\lambda^2 - \phi_1\lambda - \phi_2 = 0.$$

- The roots are easily found as are the conditions for $|\lambda| < 1$.

AR(2) Processes: 7

- One can introduce a change of variable, $z = \frac{1}{\lambda}$ to give
- $$\phi_2 z^2 + \phi_1 z - 1 = 0.$$
- The left side of this equation is called the autoregressive polynomial.
- Stationarity of the AR(2) process is equivalent to the eigenvalues of $\mathbf{M}$ having modulus strictly less than 1. This is equivalent to the roots of the autoregressive polynomial having modulus strictly greater than 1.
- This concept of the magnitude of the eigenvalues being less than 1 and the roots of the autoregressive polynomial having magnitude greater than 1 extends to AR(p) processes.

AR(2) Processes: 8

- For an AR(2) process, the autoregressive polynomial is a quadratic
$$1 - \phi_1 z - \phi_2 z^2 = 0.$$
- The roots are
$$\frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}.$$
- These roots are called the characteristic roots of the AR(2) model.
- For weak stationarity, we need the modulus of these roots to be larger than 1.